

Executive Summary

This audit report was prepared by Quantstamp, the leader in blockchain security.

Type	Strategy Vault	Documentation quality	High	
Timeline	2025-06-16 through 2025-06-25	Test quality	High	
Language	Rust	Total Findings	12	Fixed: 6 Acknowledged: 6
Methods	Architecture Review, Unit Testing, Functional Testing, Computer-Aided Verification, Manual Review	High severity findings ⓘ	1	Fixed: 1
Specification	Official Documentation 	Medium severity findings ⓘ	1	Fixed: 1
Source Code	neutral-trade/bundlev3-audit  #746151e 	Low severity findings ⓘ	7	Fixed: 4 Acknowledged: 3
Auditors	- Mostafa Yassin Auditing Engineer - Hytham Farah Auditing Engineer - Yamen Merhi Auditing Engineer	Undetermined severity findings ⓘ	0	
		Informational findings ⓘ	3	Acknowledged: 3

Summary of Findings

The protocol is a Solana-based investment-management system that lets users deposit funds into bundles (vaults), each having distinct strategies. The depositors pay an ongoing management fee plus a performance fee calculated on a high-water-mark basis based on the profit they make.

The audit uncovered 1 High-severity issue [NET-1](#), and 1 Medium issue [NET-2](#). The project ships with a solid test suite covering core flows, and all tests passed successfully. Addressing the remaining findings will further strengthen Neutral Trade’s robustness and user experience.

Fix Review Update: The high severity issue [NET-1](#) is fixed. All other findings are either resolved or acknowledged as acceptable given the protocol’s architecture.

ID	DESCRIPTION	SEVERITY	STATUS
NET-1	Refund Calculation Uses Stale Share-Mint Total	• High ⓘ	Fixed
NET-2	Precision Loss Due to Division Before Multiplication	• Medium ⓘ	Fixed
NET-3	No Slippage Tolerance for Withdrawals	• Low ⓘ	Acknowledged
NET-4	Strategy PDA Cannot Be Closed, Resulting in Locked Rent	• Low ⓘ	Acknowledged
NET-5	Enabling a Strategy without Prior Removal Can Inflate Bundle Allocation	• Low ⓘ	Fixed
NET-6	Updating Allocation of a Disabled Strategy Breaks Bundle Accounting	• Low ⓘ	Fixed
NET-7	Refill Uses Stale Oracle Data if Feed Is Out-of-date	• Low ⓘ	Fixed

ID	DESCRIPTION	SEVERITY	STATUS
NET-8	Distribution Can Be Ended Before All Allocations Are Sent	• Low ⓘ	Acknowledged
NET-9	Code Improvement	• Low ⓘ	Fixed
NET-10	Oracle Buffer of 100 % Creates Asymmetric Range—Unlimited Downside, Capped Upside	• Informational ⓘ	Acknowledged
NET-11	Withdrawal Fee Determined at Processing Time	• Informational ⓘ	Acknowledged
NET-12	Withdrawal Cooldown Resets for All Shares After Each New Deposit	• Informational ⓘ	Acknowledged

Assessment Breakdown

Quantstamp's objective was to evaluate the repository for security-related issues, code quality, and adherence to specification and best practices.

i **Disclaimer**

Only features that are contained within the repositories at the commit hashes specified on the front page of the report are within the scope of the audit and fix review. All features added in future revisions of the code are excluded from consideration in this report.

Possible issues we looked for included (but are not limited to):

- Transaction-ordering dependence
- Timestamp dependence
- Mishandled exceptions and call stack limits
- Unsafe external calls
- Integer overflow / underflow
- Number rounding errors
- Reentrancy and cross-function vulnerabilities
- Denial of service / logical oversights
- Access control
- Centralization of power
- Business logic contradicting the specification
- Code clones, functionality duplication
- Gas usage
- Arbitrary token minting

Methodology

1. Code review that includes the following
 1. Review of the specifications, sources, and instructions provided to Quantstamp to make sure we understand the size, scope, and functionality of the smart contract.
 2. Manual review of code, which is the process of reading source code line-by-line in an attempt to identify potential vulnerabilities.
 3. Comparison to specification, which is the process of checking whether the code does what the specifications, sources, and instructions provided to Quantstamp describe.
2. Testing and automated analysis that includes the following:
 1. Test coverage analysis, which is the process of determining whether the test cases are actually covering the code and how much code is exercised when we run those test cases.
 2. Symbolic execution, which is analyzing a program to determine what inputs cause each part of a program to execute.
3. Best practices review, which is a review of the smart contracts to improve efficiency, effectiveness, clarity, maintainability, security, and control based on the established industry and academic practices, recommendations, and research.
4. Specific, itemized, and actionable recommendations to help you take steps to secure your smart contracts.

Scope

The offchain components were out of scope.

Files Included

Repo: `https://github.com/neutral-trade/bundlev3-audit`

Included Paths: `programs/ntbundle/`

Operational Considerations

- The net shares minted are calculated using the `deposit_fee` in effect at execution time. Because the manager can update this parameter instantly, they could front-run deposits by raising the fee and capturing the difference. We assume the manager will act in users’ best interests.
- The same keeper address is expected to administer multiple bundles. It is expected that this key will be managed in a secure way. Alternatively, allocate a distinct keeper for each bundle to avoid all bundles compromise if a key is compromised.
- Deposits, withdrawals, and distributions all hinge on the keeper running batch instructions in the correct order and within set time windows. It is assumed that the keeper will be actively handling the protocol's operation.
- The bundle’s reported equity is the sum of its balance and the `average_external_equity` value pushed by the oracle. If this oracle lags or reports erroneous data, it can affect share prices, fees, deposits and withdrawals. We assume the oracle will update continuously and accurately.

Key Actors And Their Capabilities

Bundle Master Admin

The **Bundle Master Admin** is the account that whitelist bundle creators and it can call the following instructions:

- `set_bundle_creator()` — Add or remove an address from the bundle creation whitelist
- `change_bundle_master_admin()` — replace the bundle master admin key

Bundle Manager

The **Bundle Manager** is the address managing the operations of its own bundle and it can call the following instructions:

- `set_oracle_buffer()` — adjust the buffer used by the bundle’s equity-price oracle
- `set_max_deposit_amount()` — set the hard cap on total assets that the bundle is allowed to accept
- `set_keeper()` — replace with a new keeper address that runs batching and distribution jobs
- `set_fees()` — change `deposit_fee`, `withdrawal_fee`, `management_fee_bps`, and `performance_fee_bps`
- `set_delays()` — set `max_deposit_amount`, `withdrawal_t_min`, `withdrawal_t_max`, and `withdrawal_curve`
- `add_strategy()` — register a new strategy receiver with its `allocation_bps`
- `remove_strategy()` — delete a strategy receiver from the bundle’s whitelist and allocation set
- `update_allocations()` — modify the `allocation_bps` of a strategy
- `enable_strategy()` — re-enable a strategy that have been removed
- `manager_withdraw()` - redeem the manager’s accrued `manager_shares` for underlying assets
- `change_manager()` - transfer the manager role to a new address

Bundle Keeper

The **Bundle Keeper** is the address monitoring and executing the time-sensitive operational instructions that batch deposits, withdrawals and capital movements and it can call the following instructions:

- `pause_deposits_withdrawals()` — open or close the request window
- `net_pending_transactions()` — move pending deposits, mint shares, mark `is_netting_done`
- `process_withdrawal()` — burn user shares and transfer assets out
- `process_deposit()` — mint user shares and transfer assets in
- `refund_deposit()` — refund user assets and burn the shares
- `start_distribution()` / `distribute_to_receivers()` / `end_distribution()` — allocate capital to strategies
- `apply_fees_to_user()` — manually charge fees on a specific user account when needed
- `update_oracle()` — push a fresh external-equity reading to the bundle’s oracle data

Users

Users interact only with their own deposits and shares and can call the following instructions:

- `request_deposit()` — transfer funds into the pending-deposit vault
- `request_withdrawal()` — queue a share redemption after the lock-up
- Receive funds via `process_withdrawal()` executed by the keeper

Findings

NET-1 Refund Calculation Uses Stale Share-Mint Total

• High ⓘ Fixed

✓ Update

Marked as "Fixed" by the client.
Addressed in: `9ac820a185586306aba2c49c67fe2ce7b243aa27` .

File(s) affected: `programs/ntbundle/src/instructions/refund_deposit.rs`

Description: The `refund_deposit()` function calculates `shares_to_revert` as `user_share_of_pending_deposits * last_total_shares_minted / PRECISION`, yet `bundle_temp_data.last_total_shares_minted` is **never reduced** after each refund.

When multiple deposits are refunded in succession, earlier refunds shrink `cumulative_pending_deposits` but leave `last_total_shares_minted` unchanged; later iterations therefore cancel too many shares.

The bundle ends with the correct underlying balance but an under-reported `total_shares`, distorting share price and subsequent accounting. The same stale value is reused in `process_deposits.rs`, compounding the mismatch whenever refunds and new deposits occur in the same cycle.

Recommendation: Decrease `bundle_temp_data.last_total_shares_minted` by `shares_to_revert` each time a refund is processed.

NET-2 Precision Loss Due to Division Before Multiplication

• Medium ⓘ Fixed

✓ Update

Marked as "Fixed" by the client.

Addressed in: `f1438d7c48b8e5accdb1fe6a4685433e2cbb8e85`.

File(s) affected: `programs/ntbundle/src/state/bundle.rs`, `programs/ntbundle/src/instructions/process_deposit.rs`, `programs/ntbundle/src/instructions/refund_deposit.rs`

Description: The current implementation for calculating management fees is susceptible to a loss of precision due to its reliance on sequential integer divisions. This will cause the protocol to undercharge users for management fees over the long term systematically.

```
let time_fraction = seconds_elapsed
    .checked_mul(bundle.asset_precision as u128)
    .ok_or(ErrorCode::MathError)?
    .checked_div(SECONDS_IN_YEAR)
    .ok_or(ErrorCode::MathError)?;

// calculate fee percentage: (time_fraction * annual_fee_rate) / 10000
let mgmt_fee_pct = time_fraction
    .checked_mul(annual_fee_rate)
    .ok_or(ErrorCode::MathError)?
    .checked_div(10000) // Convert from bps to decimal
    .ok_or(ErrorCode::MathError)?;
```

The calculation is performed in three steps:

A `time_fraction` is calculated, representing the portion of a year that has passed since the last fee collection. This calculation involves a division that truncates any remainder. The `time_fraction` is then used to calculate a `mgmt_fee_pct`, which involves another division and potential truncation.

Finally, this percentage is used to determine the `management_fee_shares`.

Because integer division in computing discards the remainder, any fractional component is lost at each step.

This error, while potentially small for any single calculation, compounds and guarantees that the final fee charged will be less than or equal to the correct amount, but never more. Over thousands of transactions and users, this results in a slow but steady leak of value from the protocol's treasury.

This is also the case for the P&L for shares:

```
let pnl_usd = pnl_per_share
    .checked_mul(user_bundle_account.shares)
    .ok_or(ErrorCode::MathError)?
    .checked_div(bundle.asset_precision as u128)
    .ok_or(ErrorCode::MathError)?;

// Step 2: Second division
let perf_fee_usd = pnl_usd
    .checked_mul(bundle.performance_fee as u128)
    .ok_or(ErrorCode::MathError)?
    .checked_div(10000)
    .ok_or(ErrorCode::MathError)?;

// Step 3: Third division
performance_fee_shares = perf_fee_usd
    .checked_mul(bundle.asset_precision as u128)
    .ok_or(ErrorCode::MathError)?;
```

```
.checked_div(share_price)
.ok_or(ErrorCode::MathError)?;
```

Recommendation: To ensure mathematical accuracy and prevent value leakage, the management fee calculation should be refactored to perform all multiplications first, followed by a single division at the end.

This preserves the maximum precision throughout the calculation.

Instead of calculating intermediate values like `time_fraction` and `mgmt_fee_pct`, the final `management_fee_shares` should be calculated directly from the primary inputs (`user_bundle_account.shares`, `seconds_elapsed`, `bundle.management_fee_bps`) in one consolidated formula.

An example might look like this:

```
let numerator = user_bundle_account.shares
    .checked_mul(seconds_elapsed)
    .ok_or(ErrorCode::MathError)?
    .checked_mul(bundle.management_fee_bps as u128)
    .ok_or(ErrorCode::MathError)?;

let denominator = (SECONDS_IN_YEAR as u128)
    .checked_mul(10000)
    .ok_or(ErrorCode::MathError)?;

management_fee_shares = numerator
    .checked_div(denominator)
    .ok_or(ErrorCode::MathError)?;
```

NET-3 No Slippage Tolerance for Withdrawals

Low ⓘ

Acknowledged

i

Update

Marked as "Acknowledged" by the client.

The client provided the following explanation:

Designed this way.

File(s) affected: `programs/ntbundle/src/instructions/process_withdrawal.rs`

Description: A user's withdrawal is not priced at the time of their request but at the time of processing. This exposes the user to potentially significant slippage, as processing can be delayed, and the share price can change in the interim.

1- When a user calls `request_withdrawal()`, the protocol only records the amount of shares (e.g., \$100) to be withdrawn. The user's shares are not burned at this time.

2- The `process_withdrawal()` instruction, which calculates the asset amount to be withdrawn (`net_redemption_amount`), can be delayed by the `withdrawal_available_timestamp` mechanism.

3- Because the share price is only calculated within `process_withdrawal`, any price drop during the delay will force the user to get less asset amount than they anticipated. This makes the true cost of a withdrawal unpredictable.

Recommendation: Allow users to specify a minimum amount out they are willing to get from the withdrawal. If, during the `process_withdraw` instruction, the asset amount they are getting is less than the minimum amount they specified, then do not process the withdrawal.

NET-4

Low ⓘ

Acknowledged

Strategy PDA Cannot Be Closed, Resulting in Locked Rent

i

Update

Marked as "Acknowledged" by the client.

File(s) affected: `programs/ntbundle/src/instructions/remove_strategy.rs`

Description: The `remove_strategy` instruction fails to close the `strategy_account` PDA after disabling it in the bundle. Instead, it only sets the allowed flag to false. This orphans the on-chain account, making the SOL paid for its rent permanently irrecoverable. Every time a strategy is "removed," the protocol leaks the capital (rent) required to maintain that strategy's account. Over time, this will lead to a significant and irreversible loss of funds for the protocol manager.

Recommendation: Add another function specifically designed to close a strategy account

NET-5

Enabling a Strategy without Prior Removal Can Inflate Bundle Allocation

• Low ⓘ Fixed

✓ Update

Marked as "Fixed" by the client.
Addressed in: `7d27a2cba284a6feabee8a14351074d65975349` .

File(s) affected: `programs/ntbundle/src/instructions/enable_strategy.rs`

Description: The protocol assumes every strategy follows a strict life-cycle: it is first added, later removed, and only then re-enabled. The `enable_strategy` instruction, however, does not verify that `strategy_account.allowed` is currently false. If a bundle manager submits an `enable_strategy` call while the strategy is still marked as allowed—whether due to a failed or out-of-order `remove_strategy` transaction—the function blindly adds `strategy_account.allocation_bps` to `bundle_account.current_allocation_bps` a second time.

This double-counting breaks the bundle’s internal accounting, and may prevent subsequent strategy operations or open unintended allocation imbalances. Although unlikely in normal use, the absence of the guard leaves room for human error or automated scripts that batch transactions without checking intermediate results.

Recommendation: Insert an explicit pre-condition that re-enabling is permitted only when the strategy is currently disabled.

NET-6

Updating Allocation of a Disabled Strategy Breaks Bundle Accounting

• Low ⓘ Fixed

✓ Update

Marked as "Fixed" by the client.
Addressed in: `e488b2d17ece998fd1e708be23a8ea52ca27ad0d` .

File(s) affected: `programs/ntbundle/src/instructions/update_allocations.rs`

Description: The `update_allocation()` function is designed to adjust the position size of an active strategy. Yet the instruction never checks that `strategy_account.allowed` is `true` before proceeding. If a bundle manager calls `update_allocation` on a strategy that has already been removed, the logic becomes harmful:

1. It subtracts the strategy’s previous `allocation_bps` from `bundle_account.current_allocation_bps` , even though that subtraction already occurred during `remove_strategy` .
2. It then adds the new allocation value and writes it to the strategy account.

The modification in the bundle's allocation causes it to drift from the correct allocation, potentially turning negative or falling below the real sum of active strategies.

Situations that can trigger the bug include:

- Managers misunderstanding the life-cycle and adjusting a removed strategy.
- Transaction ordering issues where `remove_strategy` fails but a queued `update_allocation` succeeds.

Recommendation: Require that the strategy is currently enabled before modifying its allocation, or set the `strategy_account.allocation_bps` to 0 in `remove_strategy()` . Additionally, underflow checks on release mode should be enabled in `cargo.toml` .

NET-7 Refill Uses Stale Oracle Data if Feed Is Out-of-date

• Low ⓘ Fixed

✓ Update

Marked as "Fixed" by the client.
Addressed in: `38ba40468fbd01b669d698fe5337e97937fa5d0c` .

File(s) affected: `programs/ntbundle/src/instructions/perform_refill.rs`

Description: The `perform_refill()` function recomputes equity by adding the refill amount to `oracle_data.average_external_equity` .

Yet it never verifies that the oracle price is fresh; if `last_update_time` is well in the past, the function records equity with outdated market data, skewing bundle accounting, share-price calculations.

Recommendation: Introduce a freshness check—e.g., revert if `Clock::get()?.unix_timestamp - oracle_data.last_update_time` exceeds 1800 seconds—mirroring the check already enforced in `net_pending_transactions()`.

NET-8

Distribution Can Be Ended Before All Allocations Are Sent

• Low ⓘ Acknowledged

i Update

Marked as "Acknowledged" by the client.
The client provided the following explanation:

Designed this way to be able to have flexibility in terms of allocation.

File(s) affected: `programs/ntbundle/src/instructions/end_distribution.rs`

Description: The `end_distribution()` function resets `left_to_distribute` and `distribution_base_amount` without verifying that the full distribution amount has been transferred to strategies.

If the keeper closes the period prematurely, any undistributed balance is silently ignored, leaving some strategies under-funded, and skewing allocation percentages across strategies.

Recommendation: Require `left_to_distribute == 0` before ending the distribution.

NET-9 Code Improvement

• Low ⓘ Fixed

✓ Update

Marked as "Fixed" by the client.
Addressed in: `cce31374effb96b266e10cec31b547e07f1a7096`.

File(s) affected: `All Files`

Description: The code can benefit from the following validations and improvements:

1. Add `withdrawal_fee` to `ChangedCoreParams` and include `cooldown_period` plus `estimated_pending_withdrawal_value` in `WithdrawalRequested`.
2. `new_manager` instruction is not needed in `change_manager()`, unless `new_manager` is required as additional signer.
3. Drop the unused `oracle_data` account from `set_max_deposit_amount()` or reference it explicitly.
4. Remove the redundant pause check in `end_distribution()`.
5. In `request_withdrawal()`, since there is an earlier check that estimated pending withdrawals value is zero, it is possible to replace `checked_add()` for `user_bundle_account.estimated_pending_withdrawal_value` with `estimated_withdrawal_amount` with direct assignment.
6. Initialize `user_bundle_account.owner` in `initialize_bundle_depositor()` instead of `request_deposit()`.
7. Delete or annotate unused helpers like `fetch_account_balance()` in `bundle.rs`.
8. Reject `Pubkey::default()` in `change_bundle_master_admin()` to prevent null assignments.
9. Require the treasury to be a valid `TokenAccount` with `mint == asset_address` in all relevant instructions.
10. Declare `treasury_account` as `Account<'info, TokenAccount>` in `request_deposit()`.
11. The `receiver_address` in `add_strategy()` should be checked against the `Pubkey::default()` to prevent null assignments.
12. The `newkeeper` in `set_keeper()` should be checked against the `Pubkey::default()` to prevent null assignments.

Recommendation: Apply the suggested changes.

NET-10

Oracle Buffer of 100 % Creates Asymmetric Range—Unlimited Downside, Capped Upside

• Informational ⓘ Acknowledged

i Update

Marked as "Acknowledged" by the client.
The client provided the following explanation:

Designed this way.

File(s) affected: `programs/ntbundle/src/instructions/update_oracle.rs`

Description: The `update_oracle()` bounds each new equity value to `current_equity = (current_equity * oracle_buffer / 100)`.

When `oracle_buffer` is set to **100 %**, the integer calculation yields:

- **Lower bound:** `0` (-100 %; equity can be reported as zero)
- **Upper bound:** `current_equity * 2` (+100 %; equity cannot exceed 2×)

The result is an **asymmetric clamp**: the keeper may drive the bundle's equity all the way to **zero**, but can never report appreciation beyond a doubling. This allows a compromised keeper to drag the current equity to zero, but never more than the double of the current equity.

Recommendation: Disallow `oracle_buffer == 100`, or treat 100 % as unbounded in both directions.

NET-11 Withdrawal Fee Determined at Processing Time

• **Informational** ⓘ **Acknowledged**

i Update

Marked as "Acknowledged" by the client.
The client provided the following explanation:

Designed this way precisely to take the most accurate fees from the user.

File(s) affected: `programs/ntbundle/src/instructions/process_withdrawal.rs`

Description: Users request withdrawals using `request_withdrawal()`, expecting the current fee schedule to apply. However, the fee is not finalised at request time; it is recalculated during `process_withdrawal()` using the live `bundle_account.withdrawal_fee`.

The fee can change between these two events, leaving the user charged more (or less) than they had accounted for when submitting the request.

Recommendation: Deduct the fee immediately upon `request_withdrawal()` so the user's cost is fixed from the outset.

NET-12 Withdrawal Cooldown Resets for All Shares After Each New Deposit

• **Informational** ⓘ **Acknowledged**

i Update

Marked as "Acknowledged" by the client.
The client provided the following explanation:

Designed this way. We will detail the documentation.

File(s) affected: `programs/ntbundle/src/instructions/request_withdrawal.rs`

Description: Each time a user's deposit is processed, `last_deposit_timestamp` is overwritten. When that user later calls `request_withdrawal()`, the cooldown is computed from this single, most-recent timestamp, meaning **all** previously unlocked shares become subject to the full delay again.

The design forces long-term holders to wait unnecessarily if they make small top-ups, and it encourages dividing assets across multiple accounts to avoid self-induced lock-ups.

Recommendation: Track cooldown information per deposit (or in time-bucketed tranches) so only the newly added shares remain subject to the delay. If maintaining a single timestamp is preferred for simplicity, document the behavior clearly so users understand that every fresh deposit restarts the withdrawal timer for their entire position.

Auditor Suggestions

S1 Enforce Deterministic Execution Order in Distribution Lifecycle

Acknowledged

i Update

Marked as "Acknowledged" by the client.
The client provided the following explanation:

Not necessary as backend will never fail to execute the correct transactions order.

File(s) affected: programs/ntbundle/src/instructions/start_distribution.rs

Description: distribute_to_receivers() correctly checks that netting has been completed, but start_distribution() does not. This allows two valid call sequences, namely, netting before distribution begins and netting after distribution begins, increasing complexity.

Recommendation: Require bundle_temp_data.is_netting_done inside start_distribution() so the flow is deterministically: net_pending_transactions() → start_distribution() → distribute_to_receivers().

S2 Clarify Oracle Update Logic and Remove Misleading Ema Log

Fixed



Update

Marked as "Fixed" by the client.
Addressed in: 71d7f6418f9efc18f1db7109b284aff0ea9526d5 .

File(s) affected: programs/ntbundle/src/instructions/update_oracle.rs

Description: The update_oracle_data() function assigns average_external_equity directly to current_equity :

```
oracle_data.average_external_equity = current_equity;
```

Yet, both variables are included in a subsequent message:

```
msg!(
    "Updated Oracle: Current Equity = {}, New EMA = {}",
    current_equity,
    oracle_data.average_external_equity,
)
```

In particular, the "New EMA" message implies smoothing that does not exist. Either implement true EMA smoothing or change the field name and log to reflect a raw value to avoid misleading maintainers.

Recommendation: If smoothing is required, compute EMA before assignment; otherwise, rename variables and adjust the log, e.g., msg! ("External equity updated to {}", current_equity); .

Definitions

- **High severity** – High-severity issues usually put a large number of users' sensitive information at risk, or are reasonably likely to lead to catastrophic impact for client's reputation or serious financial implications for client and users.
- **Medium severity** – Medium-severity issues tend to put a subset of users' sensitive information at risk, would be detrimental for the client's reputation if exploited, or are reasonably likely to lead to moderate financial impact.
- **Low severity** – The risk is relatively small and could not be exploited on a recurring basis, or is a risk that the client has indicated is low impact in view of the client's business circumstances.
- **Informational** – The issue does not pose an immediate risk, but is relevant to security best practices or Defence in Depth.
- **Undetermined** – The impact of the issue is uncertain.
- **Fixed** – Adjusted program implementation, requirements or constraints to eliminate the risk.
- **Mitigated** – Implemented actions to minimize the impact or likelihood of the risk.
- **Acknowledged** – The issue remains in the code but is a result of an intentional business or design decision. As such, it is supposed to be addressed outside the programmatic means, such as: 1) comments, documentation, README, FAQ; 2) business processes; 3) analyses showing that the issue shall have no negative consequences in practice (e.g., gas analysis, deployment settings).

Appendix

File Signatures

The following are the SHA-256 hashes of the reviewed files. A file with a different SHA-256 hash has been modified, intentionally or otherwise, after the security review. You are cautioned that a different SHA-256 hash could be (but is not necessarily) an indication of a changed condition or potential vulnerability that was not within the scope of the review.

Files

- Repo: <https://github.com/neutral-trade/bundlev3-audit>
- [032...6ad](#) ./programs/ntbundle/Cargo.lock
 - [573...2c5](#) ./programs/ntbundle/Cargo.toml
 - [815...629](#) ./programs/ntbundle/Xargo.toml
 - [ede...c60](#) ./programs/ntbundle/src/constants.rs
 - [466...c72](#) ./programs/ntbundle/src/errors.rs
 - [59d...529](#) ./programs/ntbundle/src/events.rs
 - [a7e...28a](#) ./programs/ntbundle/src/instructions/add_strategy.rs
 - [020...cc1](#) ./programs/ntbundle/src/instructions/apply_fees_to_user.rs
 - [d94...f1a](#) ./programs/ntbundle/src/instructions/change_bundle_master_admin.rs
 - [cac...013](#) ./programs/ntbundle/src/instructions/change_manager.rs
 - [367...059](#) ./programs/ntbundle/src/instructions/distribute_to_receivers.rs
 - [d61...f16](#) ./programs/ntbundle/src/instructions/enable_strategy.rs
 - [4fb...36e](#) ./programs/ntbundle/src/instructions/end_distribution.rs
 - [d0a...427](#) ./programs/ntbundle/src/instructions/initialize_bundle.rs
 - [bf3...36d](#) ./programs/ntbundle/src/instructions/initialize_bundle_depositor.rs
 - [4b2...e8d](#) ./programs/ntbundle/src/instructions/initialize_bundle_master_account.rs
 - [7a6...5fd](#) ./programs/ntbundle/src/instructions/manager_withdraw.rs
 - [77f...cd8](#) ./programs/ntbundle/src/instructions/mod.rs
 - [c92...a79](#) ./programs/ntbundle/src/instructions/net_pending_transactions.rs
 - [1e7...fad](#) ./programs/ntbundle/src/instructions/pause_deposits_withdrawals.rs
 - [ef1...499](#) ./programs/ntbundle/src/instructions/perform_refill.rs
 - [0a9...a04](#) ./programs/ntbundle/src/instructions/process_deposit.rs
 - [a55...506](#) ./programs/ntbundle/src/instructions/process_withdrawal.rs
 - [67b...8af](#) ./programs/ntbundle/src/instructions/refund_deposit.rs
 - [0d9...51c](#) ./programs/ntbundle/src/instructions/remove_strategy.rs
 - [b64...4df](#) ./programs/ntbundle/src/instructions/request_deposit.rs
 - [644...230](#) ./programs/ntbundle/src/instructions/request_withdrawal.rs
 - [545...7e0](#) ./programs/ntbundle/src/instructions/set_bundle_creator.rs
 - [bbd...dd0](#) ./programs/ntbundle/src/instructions/set_delays.rs
 - [124...dae](#) ./programs/ntbundle/src/instructions/set_fees.rs
 - [39c...923](#) ./programs/ntbundle/src/instructions/set_keeper.rs
 - [c95...a29](#) ./programs/ntbundle/src/instructions/set_max_deposit_amount.rs
 - [c2f...71f](#) ./programs/ntbundle/src/instructions/set_oracle_buffer.rs
 - [341...43f](#) ./programs/ntbundle/src/instructions/start_distribution.rs
 - [4d1...0c4](#) ./programs/ntbundle/src/instructions/update_allocations.rs
 - [4a4...e5c](#) ./programs/ntbundle/src/instructions/update_oracle.rs
 - [f07...819](#) ./programs/ntbundle/src/lib.rs
 - [552...fa2](#) ./programs/ntbundle/src/state/bundle.rs
 - [48f...340](#) ./programs/ntbundle/src/state/mod.rs

Test Suite Results

The test data was gathered using `anchor test` . All 175 tests executed successfully.

Fix Review Update: No additional changes have been made to the test suite.

ntbundle

Sol transfered to test accounts

USDC mint created

Test accounts token accounts created

USDC minted to wallets

MasterAccount admin : 6jfNcZ1bpeddPLUcsfnauebk5hLGBuJrYLhuPcBcdTPJ

✓ Main Test Case – Step 1: Initialize master bundle account (415ms)

✓ Main Test Case – Step 2: Set bundle creator (408ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 0
Bundle underlying balance: 0
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 0
Oracle buffer: 70
Total shares: 0
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Main Test Case – Step 3: Initialize bundle with parameters (417ms)

--- Strategy Account ---

Receiver address: 34scLoUJLKx37Fm6nJX5kvhQaBMVNaYy5Lq22MB9djCm
Allocation (bps): 7500
Allowed: true

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 0
Bundle underlying balance: 0
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 7500
Oracle buffer: 70
Total shares: 0
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Main Test Case – Step 4: Add Receiver #1 to the bundle with 75% of allocation (413ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 0

Bundle underlying balance: 0
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 0
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Strategy Account ---

Receiver address: DP4rCAR6mbRANV1eLAKaxGzZCcWx5yUGJmDbBDgxLQsL
Allocation (bps): 2500
Allowed: true

✓ Main Test Case – Step 5: Add Receiver #2 to the bundle with 25% of allocation (414ms)

--- User bundle account ---

Owner AfjJBZ2o2Rjhkwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 0
Shares 0
Pending deposit 49500000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 0
Last management fee timestamp 0

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 0
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0
Treasury USDC balance : 500000000

✓ Main Test Case – Step 6: Wallet #1 requests to deposit 50 USDC (827ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 0
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 7: Pause deposit and withdrawal (397ms)

--- Oracle Data ---

Average external equity: 0
Last update time: 1750790731

✓ Main Test Case – Step 8: Updates oracle to 0 USDC (413ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 0
Bundle underlying balance: 49500000000
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 49500000000
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 49500000000
Last netting timestamp: 1750790732
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 9: Nets pending transactions (418ms)

--- User bundle account ---

Owner AfjJBZ2o2Rjhw6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 1750790732
Shares 49500000000
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1000000000
Last management fee timestamp 1750790732

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 49500000000
Last netting timestamp: 1750790732
Is netting done: true

Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 10: Processes user deposit (406ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 49500000000
Last netting timestamp: 1750790732
Is netting done: true
Distribution base amount: 49500000000
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 49500000000

✓ Main Test Case – Step 11: Starts distribution (403ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 49500000000
Last netting timestamp: 1750790732
Is netting done: true
Distribution base amount: 49500000000
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 12375000000

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 1
Bundle underlying balance: 12375000000
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 49500000000
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyjYAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Oracle Data ---

Average external equity: 37125000000
Last update time: 1750790733

✓ Main Test Case – Step 12: Distributes to receiver #1 (416ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000

Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 49500000000
Last netting timestamp: 1750790732
Is netting done: true
Distribution base amount: 49500000000
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 0

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 0
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 49500000000
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Oracle Data ---

Average external equity: 49500000000
Last update time: 1750790733

✓ Main Test Case – Step 13: Distributes to receiver #2 (412ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 49500000000
Last netting timestamp: 1750790732
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 14: Ends distribution (406ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790732
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 15: Unpause deposit and withdrawal (410ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790732
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

--- User bundle account ---

Owner AfjJBZ2o2RjHkwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 1750790732
Shares 49500000000
Pending deposit 49500000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 10000000000
Last management fee timestamp 1750790732
Treasury USDC balance : 10000000000

✓ Main Test Case – Step 16: Wallet #1 requests deposit of 50 USDC (422ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790732
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 17: Pause deposit and withdrawal again (407ms)

--- Oracle Data ---

Average external equity: 80000000000
Last update time: 1750790735

✓ Main Test Case – Step 18: Updates oracle with 30 USDC profit (413ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 49500000000
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200

Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 80128125003
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyjYAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 30628125003
Last netting timestamp: 1750790736
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 19: Nets pending transactions (411ms)

--- User bundle account ---

Owner AfjJBZ2o2RjHkwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 1750790736
Shares 76353749905
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1616161616
Last management fee timestamp 1750790736

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 30628125003
Last netting timestamp: 1750790736
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 20: Processes Wallet #1's deposit request (412ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790736
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 21: Unpauses deposit and withdrawal (no distribution) (401ms)

--- User bundle account ---

Owner AfjJBZ2o2RjHkwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 1750790736
Shares 76353749905
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1616161616
Last management fee timestamp 1750790737

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 49500000000
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 3774375098
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 80128125003
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Main Test Case – Step 22: Charges Wallet #1 management & performance fees (411ms)

--- User bundle account ---

Owner AfjJBZ2o2RjHkwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 1750790736
Shares 76353749905
Pending deposit 9900000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1616161616
Last management fee timestamp 1750790737

--- Bundle Temp Data ---

Cumulative pending deposits: 9900000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790736
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 23: Wallet #1 requests deposits of 10 USDC (830ms)

--- User bundle account ---

Owner GhNNvjiFqkhpbTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 0
Shares 0
Pending deposit 19800000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 0
Last management fee timestamp 0

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 2
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790736
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 24: **Wallet** #2 requests deposits of 20 USDC (821ms)

--- User bundle account ---

Owner GkCH1qhTxvC6EhDaaHd4Xnpv4DvwVhKCmXu4o7mc8r7p
Last deposit time 0
Shares 0
Pending deposit 19800000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 0
Last management fee timestamp 0

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 3
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790736
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 25: **Wallet** #3 requests deposits of 20 USDC (825ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 3
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790736
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 25: Pauses deposit and withdrawal again (406ms)

--- Oracle Data ---

Average external equity: 55000000000

Last update time: 1750790740

✓ Main Test Case – Step 26: Updates oracle after 25 USDC loss (411ms)

--- Bundle Account ---

Name: BUNDLE

Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry

Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB

Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt

Allocated receivers: 2

Bundle underlying balance: 99000000000

Max deposit amount: 500000000000

Withdrawal delay: 0

Withdrawal tmax: 30

Withdrawal tmin: 0

Withdrawal curve: 2

Performance fee (bps): 2000

Management fee (bps): 200

Deposit fee (bps): 100

Withdrawal fee (bps): 100

Manager shares: 3774375098

Current allocation (bps): 10000

Oracle buffer: 70

Total shares: 118083552639

Asset precision: 1000000000

Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ

Asset decimals: 9

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000

Pending deposits counter: 3

Pending withdrawals counter: 0

Pending locked withdrawals counter: 0

Last total shares minted: 37955427636

Last netting timestamp: 1750790741

Is netting done: true

Distribution base amount: 0

Is distribution started: false

Paused deposits/withdrawals: true

Left to distribute: 0

✓ Main Test Case – Step 27: Performs netting again (after loss) (408ms)

--- User bundle account ---

Owner AfjJBZ2o2Rjkhwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ

Last deposit time 1750790741

Shares 83944835280

Pending deposit 0

Pending shares 0

Estimated pending withdrawal value 0

Withdrawal available timestamp 0

Last withdrawal process timestamp 0

Last high water mark 0

Hwm per share 1587947598

Last management fee timestamp 1750790741

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000

Pending deposits counter: 2

Pending withdrawals counter: 0

Pending locked withdrawals counter: 0

Last total shares minted: 37955427636

Last netting timestamp: 1750790741
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 28: Processes Wallet #1 deposit (412ms)

--- User bundle account ---

Owner GhNNvjiFqkhpTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 1750790742
Shares 15182171054
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1304161304
Last management fee timestamp 1750790742

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 37955427636
Last netting timestamp: 1750790741
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 29: Processes Wallet #2 deposit (410ms)

--- User bundle account ---

Owner GkCH1qhTxvC6EhDaaHd4Xnpv4DvwVhKCmXu4o7mc8r7p
Last deposit time 1750790742
Shares 15182171054
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1304161304
Last management fee timestamp 1750790742

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 37955427636
Last netting timestamp: 1750790741
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 30: Processes Wallet #3 deposit (413ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000

Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 37955427636
Last netting timestamp: 1750790741
Is netting done: true
Distribution base amount: 99000000000
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 99000000000

✓ Main Test Case – Step 31: Starts new distribution round (405ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 37955427636
Last netting timestamp: 1750790741
Is netting done: true
Distribution base amount: 99000000000
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 24750000000

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 1
Bundle underlying balance: 24750000000
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 3774375250
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 118083552639
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Oracle Data ---

Average external equity: 129250000000
Last update time: 1750790743

✓ Main Test Case – Step 32: Distributes to Receiver #1 (412ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 37955427636
Last netting timestamp: 1750790741
Is netting done: true
Distribution base amount: 99000000000
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 0

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 0
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 3774375250
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 118083552639
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyjJyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Oracle Data ---

Average external equity: 154000000000
Last update time: 1750790743

✓ Main Test Case – Step 33: Distributes to Receiver #2 (409ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 49500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 37955427636
Last netting timestamp: 1750790741
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 34: Ends distribution (404ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 35: Unpauses deposit and withdrawal after distribution (416ms)

Current timestamp : 1750790745

--- User bundle account ---

Owner AfjJBZ2o2Rjhkw6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 1750790741
Shares 83944835280

Pending deposit 0
Pending shares 10000000000
Estimated pending withdrawal value 13041613040
Withdrawal available timestamp 1750790745
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1587947598
Last management fee timestamp 1750790741

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 36: **Wallet** #1 requests withdrawal of 10 shares (415ms)

Current timestamp : 1750790745

--- User bundle account ---

Owner GhNNvjiFqkhpTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 1750790742
Shares 15182171054
Pending deposit 0
Pending shares 7591085527
Estimated pending withdrawal value 9900000000
Withdrawal available timestamp 1750790746
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1304161304
Last management fee timestamp 1750790742

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 2
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 37: **Wallet** #2 requests withdrawal of 50% of the shares (415ms)

Current timestamp : 1750790746

--- User bundle account ---

Owner GkCH1qhTxvC6EhDaaHd4Xnpv4DvwVhKcMXu4o7mc8r7p
Last deposit time 1750790742
Shares 15182171054
Pending deposit 0
Pending shares 7591085527
Estimated pending withdrawal value 9900000000
Withdrawal available timestamp 1750790746
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1304161304
Last management fee timestamp 1750790742

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 3
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 38: Wallet #3 requests withdrawal of 50% of the shares (412ms)

--- User bundle account ---

Owner FSEPuC126oABqj9oeVgAqqdXvSa2mGZ3uHuHk33ePFSJ
Last deposit time 0
Shares 0
Pending deposit 9900000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 0
Last management fee timestamp 0

--- Bundle Temp Data ---

Cumulative pending deposits: 9900000000
Pending deposits counter: 1
Pending withdrawals counter: 3
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 39: Wallet #4 requests deposits of 10 USDC (821ms)

--- User bundle account ---

Owner 5kkv2TJtUi32jo6WwkFTtSDWzn6FdWVKfhTNyKtVhmky
Last deposit time 0
Shares 0
Pending deposit 14850000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 0
Last management fee timestamp 0

--- Bundle Temp Data ---

Cumulative pending deposits: 24750000000
Pending deposits counter: 2
Pending withdrawals counter: 3
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false

Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 40: Wallet #5 requests deposits of 15 USDC (821ms)

--- User bundle account ---

Owner 74UtyfxRTQYGqkBNPXAjnZGzqg29vXGzBoUkWFm4t4Eg
Last deposit time 0
Shares 0
Pending deposit 4950000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 0
Last management fee timestamp 0

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 3
Pending withdrawals counter: 3
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Main Test Case – Step 41: Wallet #6 requests deposits of 5 USDC (820ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 3
Pending withdrawals counter: 3
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790741
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 38: Pauses deposits and withdrawals again (412ms)

--- Oracle Data ---

Average external equity: 230000000000
Last update time: 1750790749

✓ Main Test Case – Step 39: Updates oracle to 230 USDC (408ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 29700000000
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000

Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 3774375250
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 133331733133
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 3
Pending withdrawals counter: 3
Pending locked withdrawals counter: 0
Last total shares minted: 15248180494
Last netting timestamp: 1750790749
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 40: Perform netting (415ms)

--- Oracle Data ---

Average external equity: 215000000000
Last update time: 1750790749

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 44700000000
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 3774375250
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 133331733133
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Main Test Case – Step 41: Receiver #1 performs refill of 15 USDC (409ms)

--- Oracle Data ---

Average external equity: 210000000000
Last update time: 1750790750

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 49700000000

Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 3774375250
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 133331733133
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Main Test Case – Step 42: Receiver #2 performs refill of 5 USDC (415ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 30222266238
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 6875918798
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 123331733133
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- User bundle account ---

Owner AfjJBZ2o2Rjhkwm6R5pwh3mKnnv4VP6Bgg8AeXVa5EtJ
Last deposit time 1750790741
Shares 70843291732
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 1750790750
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790750

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 3
Pending withdrawals counter: 2
Pending locked withdrawals counter: 0
Last total shares minted: 15248180494
Last netting timestamp: 1750790749
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0
Treasury USDC balance : 1994777337

✓ Main Test Case – Step 43: Processes Wallet #1's withdrawal request (414ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 15436551952
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 7879262350
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 115740647606
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- User bundle account ---

Owner GhNNvjiFqkhpTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 1750790742
Shares 6587741975
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 1750790751
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790751

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 3
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 15248180494
Last netting timestamp: 1750790749
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0
Treasury USDC balance : 2142634479

✓ Main Test Case – Step 44: Processes Wallet #2's withdrawal request (405ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2

Performance fee (bps): 2000
Management fee (bps): 200
Deposit fee (bps): 100
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- User bundle account ---

Owner GkCH1qhTxvC6EhDaaHd4Xnpv4DvwVhKCmXu4o7mc8r7p
Last deposit time 1750790742
Shares 6587741975
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 1750790751
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790751

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 3
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 15248180494
Last netting timestamp: 1750790749
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0
Treasury USDC balance : 2290491621

✓ Main Test Case – Step 45: Processes Wallet #3's withdrawal request (409ms)

--- User bundle account ---

Owner FSEPuC126oABqj9oeVgAqqdXvSa2mGZ3uHuHk33ePFSJ
Last deposit time 1750790751
Shares 5082726826
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790751

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 2
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 15248180494
Last netting timestamp: 1750790749
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 46: Processes Wallet #4's deposit request (404ms)

--- User bundle account ---

Owner 5kkv2TJtUi32jo6WwkFTtSDWzn6FdWVKfhTNyKtVhmky
Last deposit time 1750790752
Shares 7624090247
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790752

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 15248180494
Last netting timestamp: 1750790749
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 47: Processes Wallet #5's deposit request (410ms)

--- User bundle account ---

Owner 74UtyfxRTQYGqkBNPXAjnZGzqg29vXGzBoUkWFm4t4Eg
Last deposit time 1750790752
Shares 2541363405
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 0
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790752

--- Bundle Temp Data ---

Cumulative pending deposits: 29700000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 15248180494
Last netting timestamp: 1750790749
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Main Test Case – Step 48: Processes Wallet #6's deposit request (415ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790749
Is netting done: false
Distribution base amount: 0
Is distribution started: false

Paused deposits/withdrawals: false
Left to distribute: 0

- ✓ Main Test Case – Step 49: Unpauses deposits and withdrawals (401ms)
- ✓ Initialize Bundle Error Case – Step 1: Fails to set creator with non-admin wallet
- ✓ Initialize Bundle Error Case – Step 2: Fails to initialize bundle with non-master bundle account (408ms)
- ✓ Initialize Bundle Error Case – Step 3: Fails to initialize with 99% deposit fee (411ms)
- ✓ Initialize Bundle Error Case – Step 4: Fails to initialize with 99% performance fee
- ✓ Initialize Bundle Error Case – Step 5: Fails to initialize with 99% management fee
- ✓ Initialize Bundle Error Case – Step 6: Fails to initialize bundle with 150% oracle buffer
- ✓ Initialize Bundle Error Case – Step 7: Fails to initialize bundle with withdrawal delay > 2 years
- ✓ Initialize Bundle Error Case – Step 8: Fails to initialize a new bundle with 99% withdrawal fees
- ✓ Initialize Bundle Error Case – Step 9: Fails to initialize a new bundle with withdrawal cooldown more than 2 years

✓ Admin Error Case – Step 1: Fails to apply fees with non-keeper wallet

✓ Admin Error Case – Step 2: Fails to change manager with non-manager wallet

New manager : PublicKey [PublicKey(GhNNvjiFqkhpBTPP47JwUdpeWrqFo74NYNjtprcox9D8)] {
 _bn: <BN: e936b55f805b63a2b238756338eb28537336f05bd1c47f3bd20d5ea98d344af7>
}

New manager : PublicKey [PublicKey(GhNNvjiFqkhpBTPP47JwUdpeWrqFo74NYNjtprcox9D8)] {
 _bn: <BN: e936b55f805b63a2b238756338eb28537336f05bd1c47f3bd20d5ea98d344af7>
}

- ✓ Admin Error Case – Step 3: Change manager to another wallet (success) (669ms)
- ✓ Admin Error Case – Step 4: Fails to set fees with excessive deposit fee
- ✓ Admin Error Case – Step 5: Fails to set fees with excessive performance fee
- ✓ Admin Error Case – Step 6: Fails to set fees with excessive management fee
- ✓ Admin Error Case – Step 7: Fails to set fees with excessive withdrawal fee

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

- ✓ Admin Error Case – Step 8: Sets valid new fee parameters (357ms)

✓ Admin Error Case – Step 9: Fails to set keeper with non-manager wallet

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfl5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 10000
Oracle buffer: 70
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Admin Error Case – Step 10: Successfully sets new keeper (398ms)

✓ Admin Error Case – Step 11: Fails to set oracle buffer with non-manager wallet

✓ Admin Error Case – Step 12: Fails to set 200% oracle buffer

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfl5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 500000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Admin Error Case – Step 13: Successfully sets 90% oracle buffer (381ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfl5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 1000000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0

Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

- ✓ Admin Error Case – Step 15: Set max deposit to a reasonable level (405ms)
- ✓ Admin Error Case – Step 16: Fails to set delays with non-manager wallet
- ✓ Admin Error Case – Step 17: Fails to set withdrawal delay over 2 years
- ✓ Admin Error Case – Step 18: Fails to set cooldown over 1 month

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 1000000000000
Withdrawal delay: 1800
Withdrawal tmax: 1800
Withdrawal tmin: 900
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

- ✓ Admin Error Case – Step 19: Successfully sets 30 min delay and cooldown (778ms)

--- Strategy Account ---

Receiver address: DP4rCAR6mbRANV1eLAKaxGzZCcWx5yUGJmDbBDgxLQsL
Allocation (bps): 2500
Allowed: false

- ✓ Strategy Error Case – Step 1: Successfully removes Strategy #2 (418ms)
- ✓ Strategy Error Case – Step 2: Fails to add Receiver #3 with 50% allocation (over 100%)
- ✓ Strategy Error Case – Step 3: Fails to add Receiver #3 with 0% allocation
- ✓ Strategy Error Case – Step 4: Fails to add Receiver #3 with non-manager wallet

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 1000000000000
Withdrawal delay: 0

Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 8500
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Strategy Account ---

Receiver address: DecMSvKuC7ad6rWgfjykHnxZnx8NDZzmy3ma9xW4fF2s
Allocation (bps): 1000
Allowed: true

- ✓ Strategy Error Case – Step 5: Successfully adds Receiver #3 with 10% allocation (369ms)
- ✓ Strategy Error Case – Step 6: Fails to update Receiver #3 allocation to 30% (over limit)
- ✓ Strategy Error Case – Step 7: Fails to update Receiver #3 allocation to 0%
- ✓ Strategy Error Case – Step 8: Fails to update Receiver #3 allocation to 20% with non-manager wallet

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 1000000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 9500
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Strategy Account ---

Receiver address: DecMSvKuC7ad6rWgfjykHnxZnx8NDZzmy3ma9xW4fF2s
Allocation (bps): 2000
Allowed: true

- ✓ Strategy Error Case – Step 9: Successfully updates Receiver #3 allocation to 20% (369ms)
- ✓ Strategy Error Case – Step 10: Fails to remove Receiver #3 with non-manager wallet

--- Strategy Account ---

Receiver address: DecMSvKuC7ad6rWgfjykHnxZnx8NDZzmy3ma9xW4fF2s
Allocation (bps): 2000
Allowed: false

- ✓ Strategy Error Case – Step 11: Successfully removes Receiver #3 (392ms)

✓ Strategy Error Case – Step 12: Fails to enable Receiver #3 with non-manager wallet

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 1000000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 9500
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Strategy Account ---

Receiver address: DecMSvKuC7ad6rWgfjyKHnxZnx8NDZzmy3ma9xW4fF2s
Allocation (bps): 2000
Allowed: true

✓ Strategy Error Case – Step 13: Successfully enables Receiver #3 (396ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 1000000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 7500
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Strategy Account ---

Receiver address: DecMSvKuC7ad6rWgfjyKHnxZnx8NDZzmy3ma9xW4fF2s
Allocation (bps): 2000
Allowed: false

✓ Strategy Error Case – Step 14: Successfully removes Receiver #3 again (409ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry

Keeper: 6XkrGpbtgZZRDWMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 650837666
Max deposit amount: 1000000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882605902
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 108149562079
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Strategy Account ---

Receiver address: DP4rCAR6mbRANV1eLAKaxGzZCcWx5yUGJmDbBDgxLQsL
Allocation (bps): 2500
Allowed: true

- ✓ Strategy Error Case – Step 15: Re-enables Receiver #2 (406ms)
- ✓ Strategy Error Case – Step 16: Fails to enable Receiver #3 again (allocation > 100%)
- ✓ Strategy Error Case – Step 17: Fails to remove Receiver #3 again (already removed)
- ✓ Request Flow – Step 1: Wallet #2 requests to deposit 0 USDC (should fail)
- ✓ Request Flow – Step 2: Wallet #2 requests to deposit 10,000 USDC (should exceed max)

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790749
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

--- User bundle account ---

Owner GhNNvjiFqkhpTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 1750790742
Shares 6587741975
Pending deposit 9500000000
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 1750790751
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790751

- ✓ Request Flow – Step 3: Wallet #2 requests to deposit 10 USDC (success) (347ms)
- ✓ Request Flow – Step 4: Wallet #2 requests another deposit (should fail due to existing pending)
- ✓ Request Flow – Step 5: Wallet #2 tries to request withdrawal while deposit pending
- ✓ Request Flow – Step 6: Wallet #1 tries to withdraw 200% of shares (should fail)

- ✓ Request Flow – Step 7: **Wallet** #1 tries to withdraw 0 shares (should fail)

--- User bundle account ---

Owner AfjJBZ2o2Rjhkwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ
Last deposit time 1750790741
Shares 70843291732
Pending deposit 0
Pending shares 35421645866
Estimated pending withdrawal value 68993338759
Withdrawal available timestamp 1750790764
Last withdrawal process timestamp 1750790750
Last high water mark 0
Hwm per share 1947773376
Last management fee timestamp 1750790750

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 1
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790749
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

- ✓ Request Flow – Step 8: **Wallet** #1 successfully withdraws 50% of shares (340ms)
- ✓ Request Flow – Step 9: **Wallet** #1 tries to deposit while withdrawal pending (should fail)
- ✓ Request Flow – Step 10: **Wallet** #1 tries to withdraw again (should fail due to existing withdrawal)
- ✓ Before Pause – Step 11: Fails netting when bundle not paused
- ✓ Before Pause – Step 12: Fails to update oracle while not paused
- ✓ Before Pause – Step 13: Fails to start distribution while not paused
- ✓ Before Pause – Step 14: Fails to process **Wallet** #1 withdrawal when not paused
- ✓ Before Pause – Step 15: Fails to process **Wallet** #2 deposit when not paused
- ✓ Before Pause – Step 16: Fails to pause with non-keeper wallet

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 1
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790749
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

- ✓ Pause Stage – Step 17: Successfully pauses deposits and withdrawals (270ms)
- ✓ Pause Stage – Step 18: Fails to unpause while transactions are pending
- ✓ Pause Stage – Step 20: Fails to update oracle with non-keeper wallet
- ✓ Pause Stage – Step 21: Fails to update oracle with huge equity
- ✓ Pause Stage – Step 22: Fails to update oracle with too small equity

--- Oracle Data ---

Average external equity: 175000000000

Last update time: 1750790761

- ✓ Pause Stage – Step 23: Successfully updates oracle with +40 USDC (352ms)
- ✓ Pause Stage – Step 24: Fails to process Wallet #1 withdrawal when netting not done
- ✓ Pause Stage – Step 25: Fails to process Wallet #2 deposit when netting not done
- ✓ Pause Stage – Step 26: Fails netting with non-keeper wallet

--- Bundle Account ---

Name: BUNDLE

Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry

Keeper: 6XkrGpbtgZZRDwMVW86F7yfl5K1DpqBCw9zTxZnb3WRB

Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt

Allocated receivers: 2

Bundle underlying balance: 10150837666

Max deposit amount: 1000000000000

Withdrawal delay: 0

Withdrawal tmax: 30

Withdrawal tmin: 0

Withdrawal curve: 2

Performance fee (bps): 1500

Management fee (bps): 500

Deposit fee (bps): 500

Withdrawal fee (bps): 100

Manager shares: 8882605902

Current allocation (bps): 10000

Oracle buffer: 90

Total shares: 113998784625

Asset precision: 1000000000

Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ

Asset decimals: 9

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000

Pending deposits counter: 1

Pending withdrawals counter: 1

Pending locked withdrawals counter: 0

Last total shares minted: 5849222546

Last netting timestamp: 1750790762

Is netting done: true

Distribution base amount: 0

Is distribution started: false

Paused deposits/withdrawals: true

Left to distribute: 0

- ✓ Pause Stage – Step 27: Successfully performs netting (359ms)
- ✓ Pause Stage – Step 28: Fails to perform netting again
- ✓ Pause Stage – Step 29: Fails to start distribution with pending deposits/withdrawals
- ✓ Pause Stage – Step 30: Fails to process Wallet #1's deposit (none exists)
- ✓ Pause Stage – Step 31: Fails to process Wallet #2's withdrawal (none exists)
- ✓ Pause Stage – Step 32: Fails to process Wallet #1 withdrawal with non-keeper
- ✓ Pause Stage – Step 33: Fails to process Wallet #2 deposit with non-keeper

Waiting 30 seconds...

- ✓ Pause Stage – Step 34: Fails to process Wallet #1 withdrawal due to insufficient bundle funds (30022ms)

✓ Pause Stage – Step 35: **Receiver** #3 tries to refill 50 USDC (not whitelisted)

✓ Pause Stage – Step 36: **Receiver** #1 tries to refill 0 USDC (invalid amount)

--- Oracle Data ---

Average external equity: 25000000000

Last update time: 1750790792

--- Bundle Account ---

Name: **BUNDLE**

Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry

Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB

Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt

Allocated receivers: 2

Bundle underlying balance: 160150837666

Max deposit amount: 1000000000000

Withdrawal delay: 0

Withdrawal tmax: 30

Withdrawal tmin: 0

Withdrawal curve: 2

Performance fee (bps): 1500

Management fee (bps): 500

Deposit fee (bps): 500

Withdrawal fee (bps): 100

Manager shares: 8882605902

Current allocation (bps): 10000

Oracle buffer: 90

Total shares: 113998784625

Asset precision: 1000000000

Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ

Asset decimals: 9

✓ Pause Stage – Step 37: **Receiver** #1 refills 150 USDC (196ms)

--- Bundle Account ---

Name: **BUNDLE**

Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry

Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB

Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt

Allocated receivers: 2

Bundle underlying balance: 102620861139

Max deposit amount: 1000000000000

Withdrawal delay: 0

Withdrawal tmax: 30

Withdrawal tmin: 0

Withdrawal curve: 2

Performance fee (bps): 1500

Management fee (bps): 500

Deposit fee (bps): 500

Withdrawal fee (bps): 100

Manager shares: 8882610577

Current allocation (bps): 10000

Oracle buffer: 90

Total shares: 78577138759

Asset precision: 1000000000

Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ

Asset decimals: 9

--- User bundle account ---

Owner AfjJBZ2o2Rjhkwm6R5pwh3mKnnv4VP6Bqg8AeXVa5EtJ

Last deposit time 1750790741

Shares 35421641191

Pending deposit 0

Pending shares 0

Estimated pending withdrawal value 0

Withdrawal available timestamp 0

Last withdrawal process timestamp 1750790792

Last high water mark 0

Hwm per share 1947773376
Last management fee timestamp 1750790792

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 1
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 5849222546
Last netting timestamp: 1750790762
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

- ✓ Pause Stage – Step 41: Processes Wallet #1 withdrawal (success) (410ms)

--- User bundle account ---

Owner GhNNvjiFqkhpBTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 1750790793
Shares 12436964087
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 1750790751
Last high water mark 0
Hwm per share 1795569033
Last management fee timestamp 1750790793

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 5849222546
Last netting timestamp: 1750790762
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

- ✓ Pause Stage – Step 42: Processes Wallet #2 deposit (success) (416ms)
- ✓ Pause Stage – Step 43: Fails to end distribution (not started)
- ✓ Pause Stage – Step 44: Fails to start distribution with non-keeper wallet
- ✓ Pause Stage – Step 45: Fails to distribute to Receiver #1 before start

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 5849222546
Last netting timestamp: 1750790762
Is netting done: true
Distribution base amount: 102620861139
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 102620861139

- ✓ Pause Stage – Step 46: Starts distribution (success) (364ms)
- ✓ Pause Stage – Step 47: Fails to start distribution again (already started)

- ✓ Pause Stage – Step 48: Fails to unpause during active distribution
- ✓ Pause Stage – Step 49: Fails to distribute to Receiver #1 with non-keeper
- ✓ Pause Stage – Step 51: Fails to distribute to Receiver #3 (not whitelisted)

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 5849222546
Last netting timestamp: 1750790762
Is netting done: true
Distribution base amount: 102620861139
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 25655215285

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 1
Bundle underlying balance: 25655215285
Max deposit amount: 1000000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611011
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 78577138759
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Oracle Data ---

Average external equity: 101965645854
Last update time: 1750790794

- ✓ Pause Stage – Step 52: Successfully distributes to Receiver #1 (346ms)
- ✓ Pause Stage – Step 53: Fails to distribute again to Receiver #1 (already allocated)

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 5849222546
Last netting timestamp: 1750790762
Is netting done: true
Distribution base amount: 102620861139
Is distribution started: true
Paused deposits/withdrawals: true
Left to distribute: 1

--- Bundle Account ---

Name: BUNDLE

Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 1
Max deposit amount: 1000000000000
Withdrawal delay: 0
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611011
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 78577138759
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Oracle Data ---

Average external equity: 127620861138
Last update time: 1750790794

- ✓ Pause Stage – Step 55: Successfully distributes to Receiver #2 (396ms)
- ✓ Pause Stage – Step 56: Fails to end distribution with non-keeper wallet

--- Bundle Temp Data ---

Cumulative pending deposits: 9500000000
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 5849222546
Last netting timestamp: 1750790762
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

- ✓ Pause Stage – Step 57: Successfully ends distribution (384ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790762
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

- ✓ Pause Stage – Step 58: Successfully unpauses the bundle (413ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 1
Max deposit amount: 1000000000000

Withdrawal delay: 3600
Withdrawal tmax: 30
Withdrawal tmin: 0
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611011
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 78577138759
Asset precision: 10000000000
Asset address: AZmSUR2DpKLyjYAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Pause Stage – Step 59: Fails to withdraw immediately due to withdrawal delay (424ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790762
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 59: Pause deposit and withdrawal (393ms)

✓ Extra Deposit & Withdrawal – Step 60: Wallet #1 cannot deposit when paused

✓ Extra Deposit & Withdrawal – Step 61: Wallet #2 cannot withdraw when paused

✓ Extra Deposit & Withdrawal – Step 62: Netting fails when there are no transactions

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790762
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 63: Unpause deposit and withdrawal (361ms)

Current timestamp : 1750790797

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 1
Max deposit amount: 10000000000000
Withdrawal delay: 1
Withdrawal tmax: 10
Withdrawal tmin: 10
Withdrawal curve: 2
Performance fee (bps): 1500

Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611011
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 78577138759
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- User bundle account ---

Owner GhNNvjiFqkhpTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 1750790793
Shares 12436964087
Pending deposit 0
Pending shares 12436964087
Estimated pending withdrawal value 20199463760
Withdrawal available timestamp 1750790807
Last withdrawal process timestamp 1750790751
Last high water mark 0
Hwm per share 1795569033
Last management fee timestamp 1750790793

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790762
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 64: Wallet #2 make a withdrawal request (828ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790762
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 65: Immediately pause deposit and withdrawal (401ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbgtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 1
Max deposit amount: 1000000000000
Withdrawal delay: 1
Withdrawal tmax: 10
Withdrawal tmin: 10
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500

Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611011
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 78577138759
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyjYAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790798
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 66: Complete netting (412ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 1
Last total shares minted: 0
Last netting timestamp: 1750790798
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

--- Oracle Data ---

Average external equity: 97620861138
Last update time: 1750790798

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 30000000001
Max deposit amount: 100000000000
Withdrawal delay: 1
Withdrawal tmax: 10
Withdrawal tmin: 10
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611011
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 78577138759
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyjYAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Extra Deposit & Withdrawal – Step 67: Wallet #2 withdrawal request is processed (823ms)

✓ Extra Deposit & Withdrawal – Step 68: [Wallet](#) #2 withdrawal cannot be processed again

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790798
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: false
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 69: [Unpause deposit and withdrawal](#) (384ms)

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790798
Is netting done: false
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 71: [Pause again before final withdrawal](#) (412ms)

--- Bundle Account ---

Name: [BUNDLE](#)
Manager: [39JadiCUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry](#)
Keeper: [6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB](#)
Treasury account: [9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt](#)
Allocated receivers: 2
Bundle underlying balance: [30000000001](#)
Max deposit amount: [100000000000](#)
Withdrawal delay: 1
Withdrawal tmax: 10
Withdrawal tmin: 10
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: [8882611011](#)
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: [78577138759](#)
Asset precision: [1000000000](#)
Asset address: [AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ](#)
Asset decimals: 9

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 1
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790800
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 72: Complete netting again (412ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 9800536766
Max deposit amount: 1000000000000
Withdrawal delay: 1
Withdrawal tmax: 10
Withdrawal tmin: 10
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611334
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 66140174995
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

--- User bundle account ---

Owner GhNNvjiFqkhpTPP47JwUdpeWrqFo74NYNjtprcox9D8
Last deposit time 1750790793
Shares 0
Pending deposit 0
Pending shares 0
Estimated pending withdrawal value 0
Withdrawal available timestamp 0
Last withdrawal process timestamp 1750790810
Last high water mark 0
Hwm per share 1795569033
Last management fee timestamp 1750790810

--- Bundle Temp Data ---

Cumulative pending deposits: 0
Pending deposits counter: 0
Pending withdrawals counter: 0
Pending locked withdrawals counter: 0
Last total shares minted: 0
Last netting timestamp: 1750790800
Is netting done: true
Distribution base amount: 0
Is distribution started: false
Paused deposits/withdrawals: true
Left to distribute: 0

✓ Extra Deposit & Withdrawal – Step 73: Final withdrawal from Wallet #2 is processed (10254ms)

✓ Manager Fee Withdrawal – Step 64: Try to withdraw with non-manager wallet

✓ Manager Fee Withdrawal – Step 65: Try to withdraw without refilling

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 29800536766
Max deposit amount: 1000000000000
Withdrawal delay: 1

Withdrawal tmax: 10
Withdrawal tmin: 10
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 8882611334
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 66140174995
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Manager Fee Withdrawal – Step 66: Receiver #1 and #2 refill enough USDC (772ms)

--- Bundle Account ---

Name: BUNDLE
Manager: 39JadICUZ4SjKznEjjBGgb812Ayg3mB8XLSfGF63oRry
Keeper: 6XkrGpbtgZZRDwMVW86F7yfL5K1DpqBCw9zTxZnb3WRB
Treasury account: 9HR7nhTeytVveM7C5mm1mahfsjZYcoNBYg1jYfJGEhDt
Allocated receivers: 2
Bundle underlying balance: 15373866039
Max deposit amount: 1000000000000
Withdrawal delay: 1
Withdrawal tmax: 10
Withdrawal tmin: 10
Withdrawal curve: 2
Performance fee (bps): 1500
Management fee (bps): 500
Deposit fee (bps): 500
Withdrawal fee (bps): 100
Manager shares: 0
Current allocation (bps): 10000
Oracle buffer: 90
Total shares: 57257563661
Asset precision: 1000000000
Asset address: AZmSUR2DpKLyijyAR8xUiWVLsS439MCMKkkaJR2B1BSQ
Asset decimals: 9

✓ Manager Fee Withdrawal – Step 67: Manager withdraws (416ms)

✓ Manager Fee Withdrawal – Step 68: Try to withdraw manager shares again (should fail)

170 passing (2m)

Changelog

- 2025-06-25 - Initial report
- 2025-07-04 - Final report

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Quantstamp’s team consists of cybersecurity experts hailing from globally recognized organizations including Microsoft, AWS, BMW, Meta, and the Ethereum Foundation. Quantstamp engineers hold PhDs or advanced computer science degrees, with decades of combined experience in formal verification, static analysis, blockchain audits, penetration testing, and original leading-edge research.

To date, Quantstamp has performed more than 500 audits and secured over \$200 billion in digital asset risk from hackers. Quantstamp has worked with a diverse range of customers, including startups, category leaders and financial institutions. Brands that Quantstamp has worked with include Ethereum 2.0, Binance, Visa, PayPal, Polygon, Avalanche, Curve, Solana, Compound, Lido, MakerDAO, Arbitrum, OpenSea and the World Economic Forum.

Quantstamp's collaborations and partnerships showcase our commitment to world-class research, development and security. We're honored to work with some of the top names in the industry and proud to secure the future of web3.

Notable Collaborations & Customers:

- Blockchains: Ethereum 2.0, Near, Flow, Avalanche, Solana, Cardano, Binance Smart Chain, Hedera Hashgraph, Tezos
- DeFi: Curve, Compound, Maker, Lido, Polygon, Arbitrum, SushiSwap
- NFT: OpenSea, Parallel, Dapper Labs, Decentraland, Sandbox, Axie Infinity, Illuvium, NBA Top Shot, Zora
- Academic institutions: National University of Singapore, MIT

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